

INF201 - AI Chat Log Summary

Week 40: VS Code, Git/GitHub and file-processing preparation

1. Git and GitHub setup

- **Problem:** The command `git` was initially not recognized in the VS Code PowerShell terminal.
- **Solution:** Git for Windows was installed and VS Code was restarted. The installation was verified with `git --version`.
- **Configuration:** Git was configured with my name and GitHub email, and new repositories were set to use `main` as the default branch.
- **GitHub workflow:** I created a GitHub repository, copied its HTTPS repository URL and used the repository from VS Code.

2. First Git workflow in VS Code

The Source Control panel in VS Code was used to understand the normal Git workflow:

Step	Meaning
Stage	Select the changed files that should be included in the next checkpoint.
Commit	Save a local checkpoint with a descriptive message, e.g. "Add hello script".
Push	Send the local commit to the repository on GitHub.

3. Python in VS Code

- VS Code initially had no usable Python interpreter, so a simple `print("hello, git")` did not run.
- Python was installed and connected to VS Code. A later terminal issue showed that Windows could not find the `python` command through `PATH`, so the `py` launcher was suggested as an alternative.
- The distinction between writing code and actually executing the file with a Python interpreter was clarified.

4. Task 2 - planned approach

Task 2 is based on a small data-processing pipeline using several file formats. The planned sequence is:

- Read settings from `config.yml`.
- Read sensor information from `sensors.xlsx` and calibration data from `calibrations.csv`.
- Match the two datasets using `sensor_id`.
- Filter sensors where `days_since_calibration` is greater than the configured maximum.
- Write the overdue sensors to a formatted JSON file.

5. Relevant Python modules

`pandas` can be used for CSV/Excel data and joining tables; `openpyxl` supports Excel files; `PyYAML` reads YAML configuration; and `json` is part of Python's standard library and is used for JSON output.

Main learning outcome: The conversation focused on understanding the workflow and tools step by step rather than receiving a complete solution.